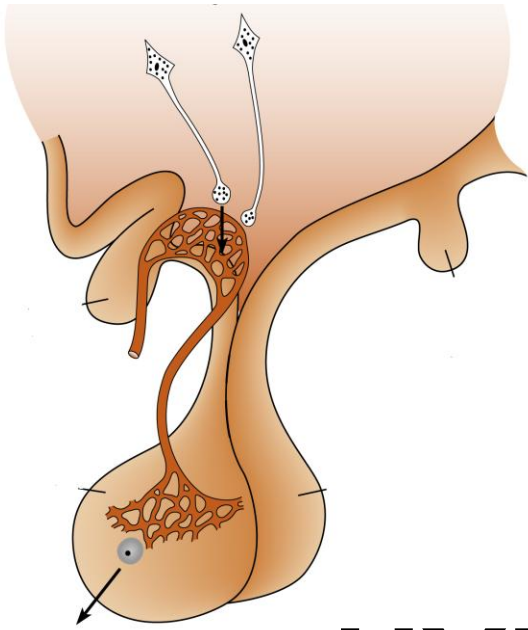
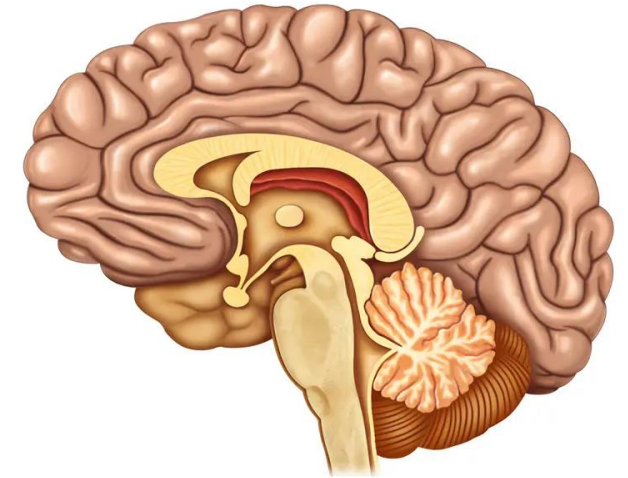




BPM 2

ER: Lecture 02



HYPOTHALAMUS & PITUITARY FUNCTIONS

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Recommended Reading

Chapter 31: Hypothalamus and Pituitary gland. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell.

Slides with blue background are the supplemental information of the previous slides and thus, might not be explained again during the lecture time. However, please read the slides for clarification as they are equally important to know

Glossary

ACTH	Adrenocorticotrophic hormone (corticotropin)
ADH	Antidiuretic hormone
AVP	Arginine-Vasopressin
SON	Supraoptic neuron
PVN	Paraventricular neuron
CRH	Corticotropin-releasing hormone
FSH	Follicle-stimulating hormone
GABA	Gamma-aminobutyric acid
GH	Growth hormone (somatotropin)
GHBP	Growth hormone–binding protein
GHRH	Growth hormone–releasing hormone
SRIF	Somatotropin release inhibiting factor (Somatostatin)
GnRH	Gonadotropin-releasing hormone

hCG	Human chorionic gonadotropin
hPL	Human placental lactogen
IGF	Insulin-like growth factor
LH	Luteinizing hormone
MSH	Melanocyte-stimulating hormone
PIH	Prolactin-inhibiting hormone (dopamine)
PRH	Prolactin-releasing hormone (TRH)
POMC	Pro-opiomelanocortin
PRL	Prolactin
SHBG	Sex hormone–binding globulin
SIADH	Syndrome of inappropriate secretion of antidiuretic hormone
TRH	Thyrotropin-releasing hormone
TSH	Thyroid-stimulating hormone (thyrotropin)

Learning Objectives

SOM.MK.II.BPM2.1.ER.1.PHYS.1003. Explain the relationship between anterior pituitary and the hypothalamus and the regulation of hypothalamic hormone release.

SOM.MK.II.BPM2.1.ER.1.PHYS.1004. Identify appropriate hypothalamic factors that control the secretion of each of the anterior pituitary hormones.

SOM.MK.II.BPM2.1.ER.1.PHYS.1005. List the different cell types that synthesize and secrete anterior pituitary hormones.

SOM.MK.II.BPM2.1.ER.1.PHYS.1140. Explain the chemical nature and clinical significance of the three different types of anterior pituitary hormones and their functional relationship with placental hormones.

SOM.MK.II.BPM2.1.ER.1.PHYS.1006. Understand negative feedback control of anterior pituitary hormone secretion at multiple levels.

SOM.MK.II.BPM2.1.ER.1.PHYS.1007. Describe growth hormone, which 2nd messenger it activates and what triggers its release.

Learning Objectives

SOM.MK.II.BPM2.1.ER.1.PHYS.1008. Describe the effects of GHRH and somatostatin on GH release and their mechanism of action on target using which 2nd messenger.

SOM.MK.II.BPM2.1.ER.1.PHYS.1009. Understand the pattern of secretion of growth hormone.

SOM.MK.II.BPM2.1.ER.1.PHYS.1010. Describe the metabolic and growth promoting actions of growth hormone. Describe their mechanism of action on target organs.

SOM.MK.II.BPM2.1.ER.1.PHYS.1011. Describe the relationship between growth hormone and the insulin-like growth factors and their binding proteins in the regulation of growth.

SOM.MK.II.BPM2.1.ER.1.PHYS.1012. Predict the developmental effects of too much or too little growth hormone.

SOM.MK.II.BPM2.1.ER.1.PHYS.1013. Evaluate the feedback mechanisms in the regulation of growth hormone levels.

SOM.MK.II.BPM2.1.ER.1.PHYS.1141. Explain additional factors that can affect growth.

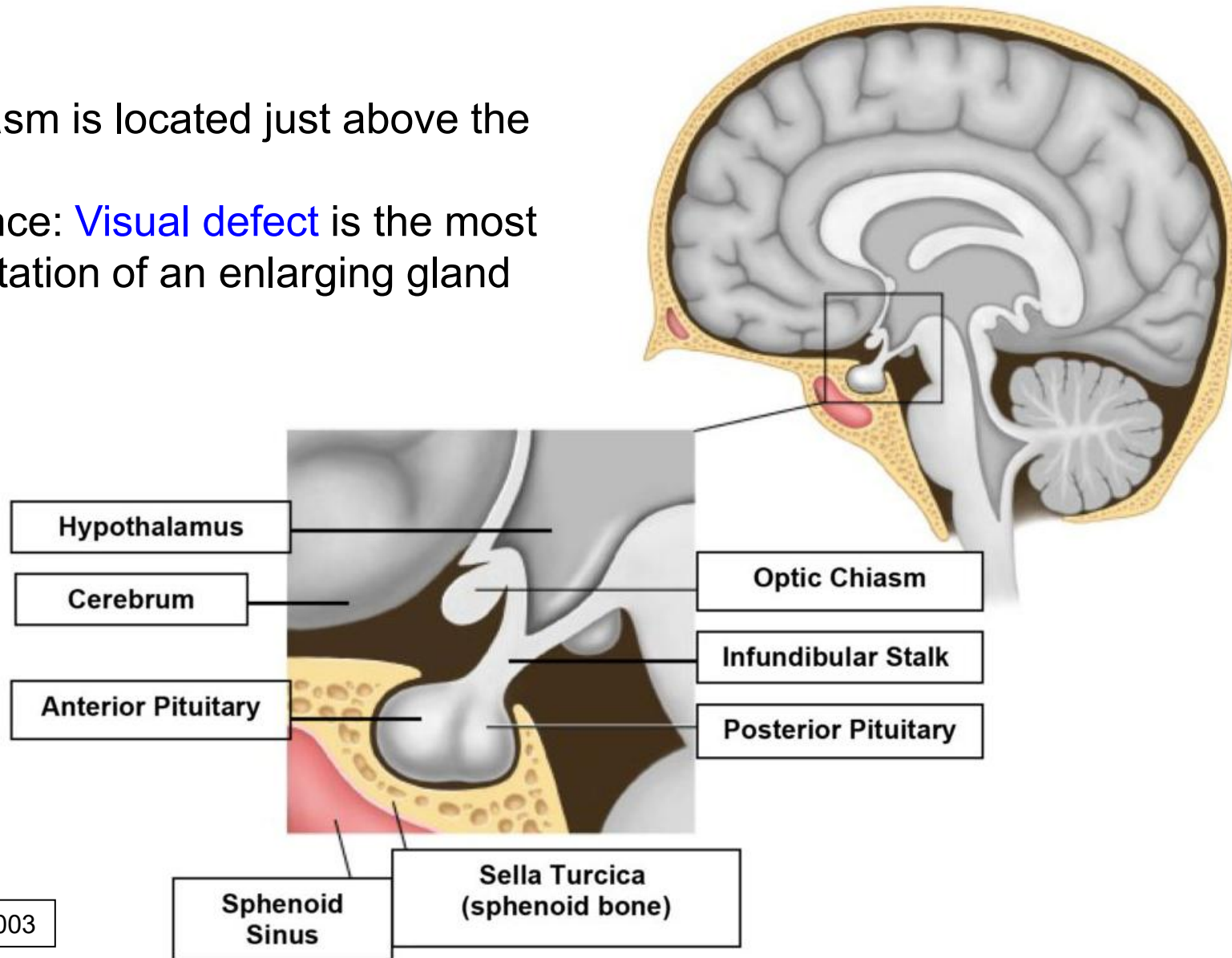
Lecture Outline

- ❖ **Relationship between hypothalamus and pituitary gland**
- ❖ **Hormones of anterior pituitary and Negative feedback regulation**
- ❖ **Human Growth Hormone: Physiology and Pathophysiology**

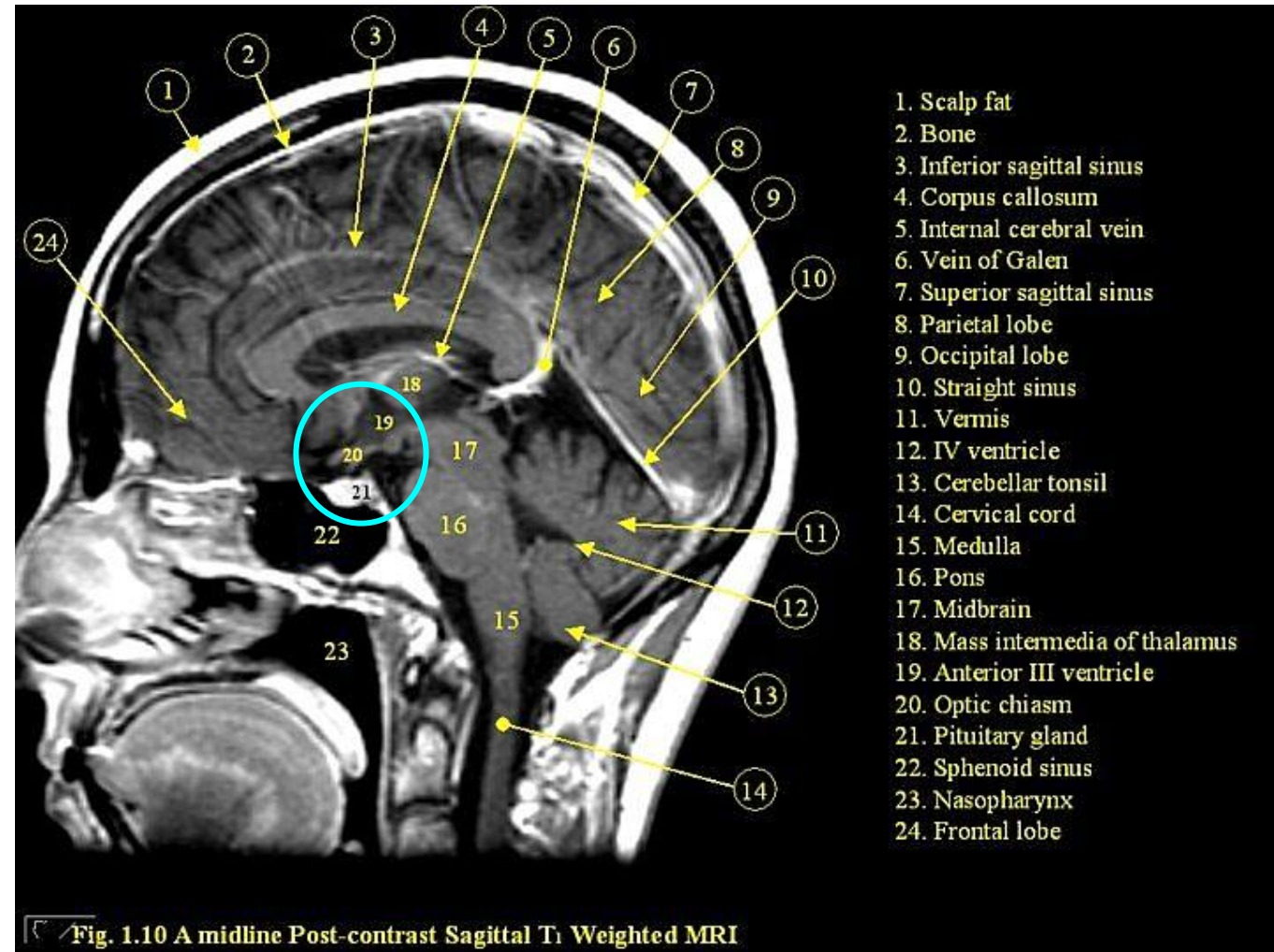
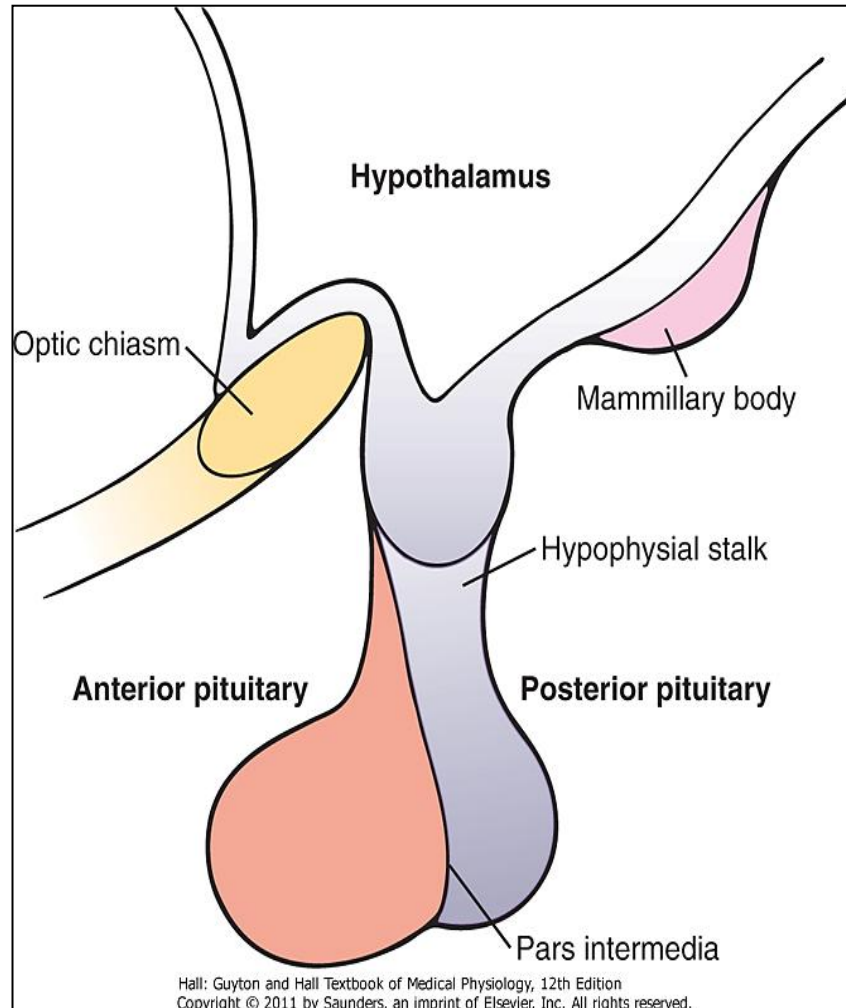
STRUCTURE OF THE HYPOTHALAMIC–PITUITARY AXIS

Note: Optic Chiasm is located just above the pituitary gland.

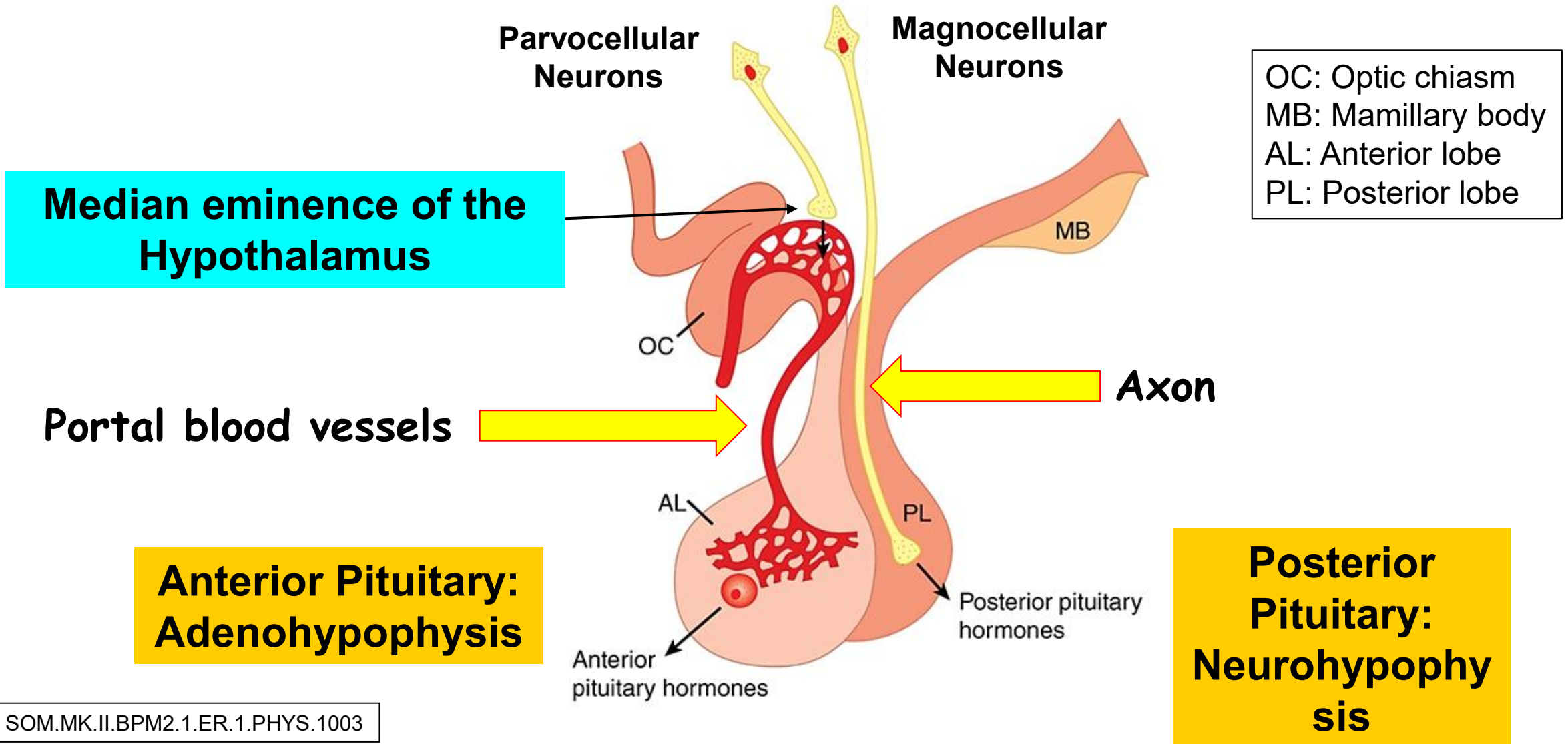
Clinical importance: **Visual defect** is the most common presentation of an enlarging gland



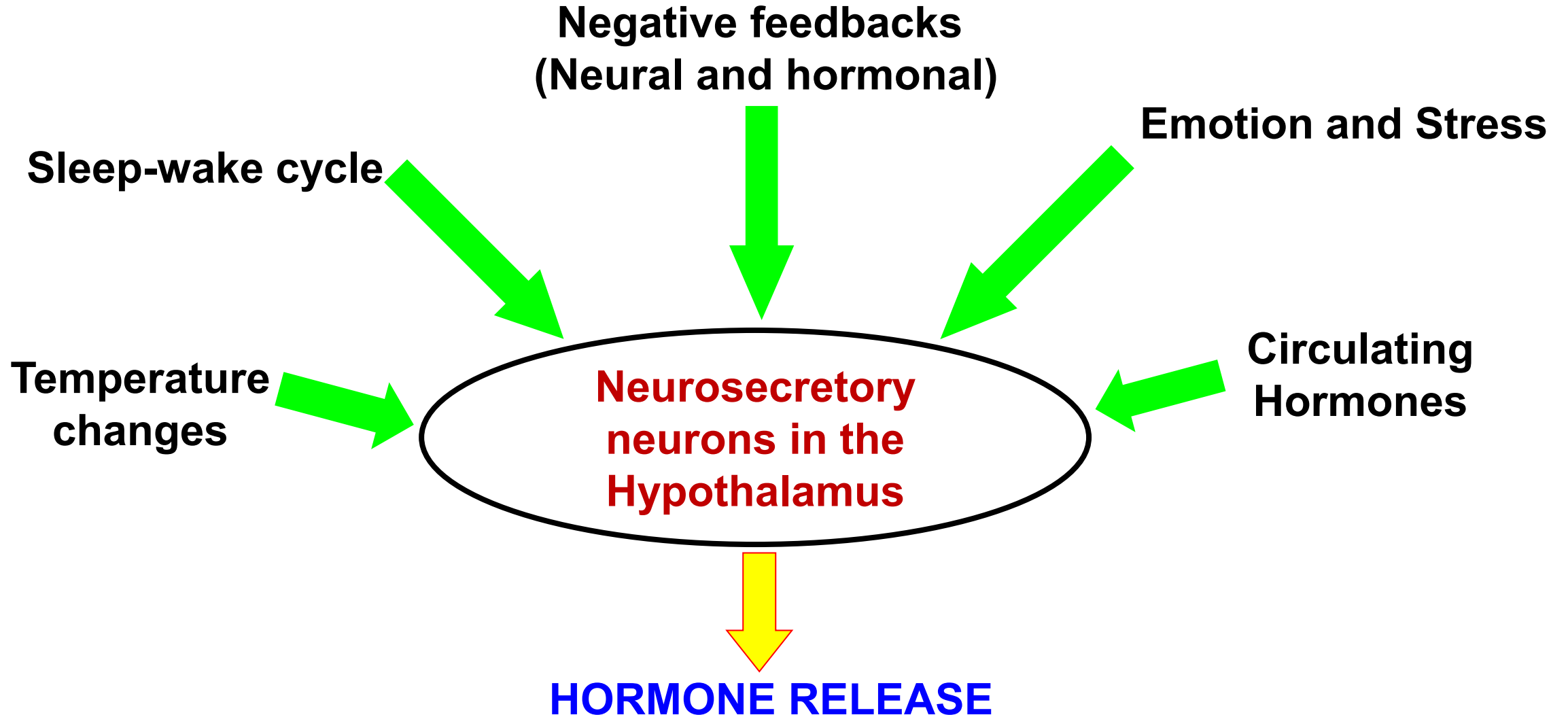
STRUCTURE OF THE HYPOTHALAMIC–PITUITARY AXIS



Hypothalamic pituitary axis



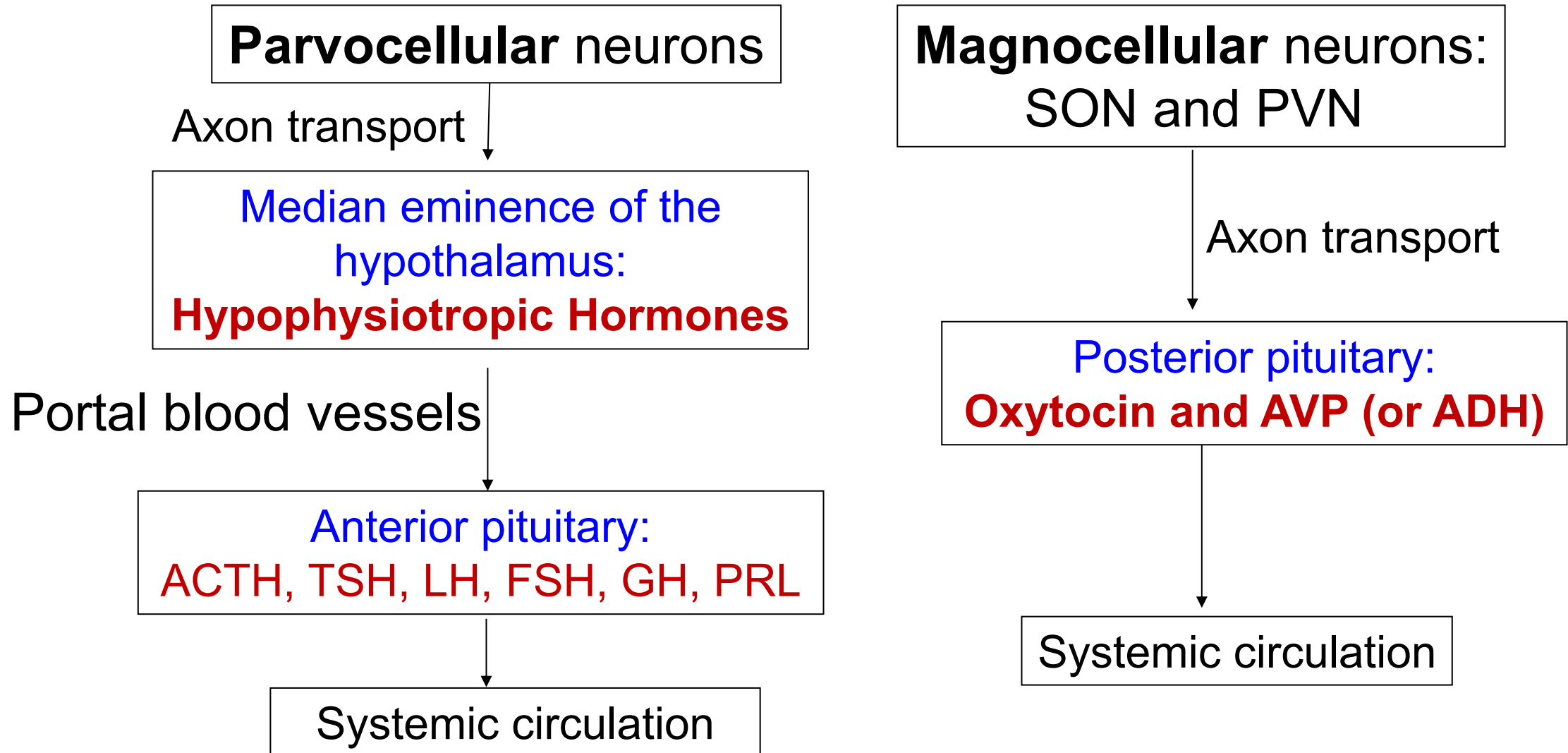
External or internal signals affect Hypothalamic Hormones



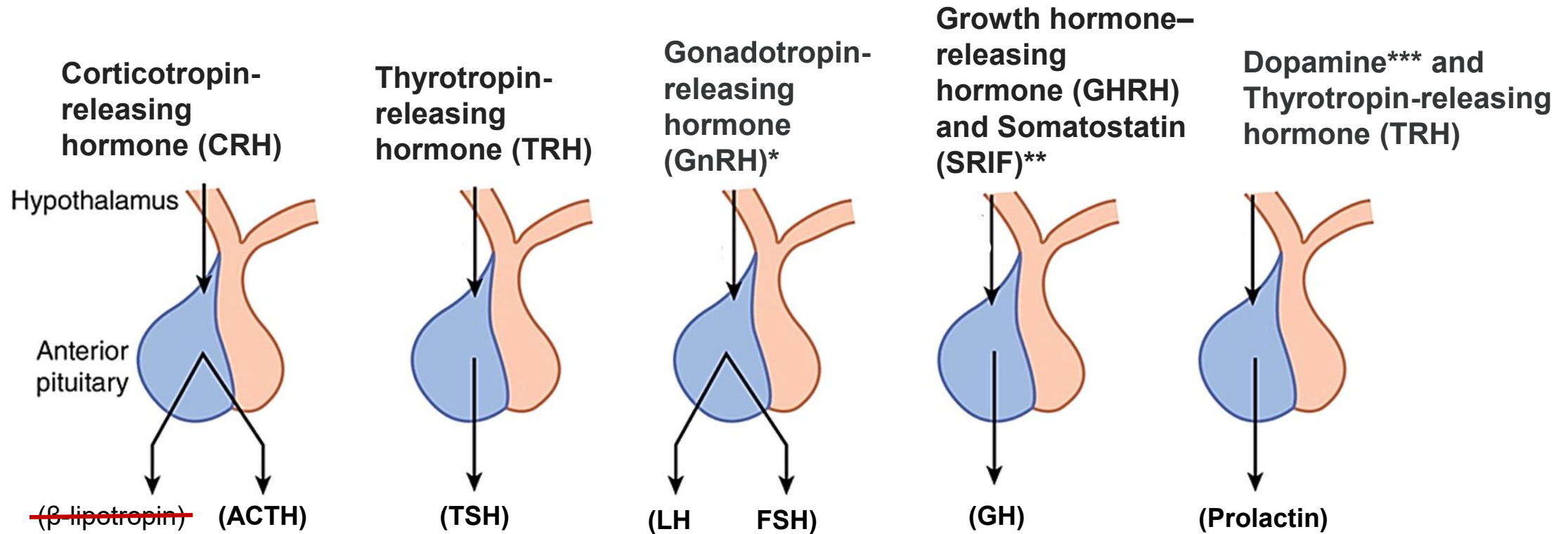
Lecture Outline

- ❖ Relationship between hypothalamus and pituitary gland
- ❖ Hormones of anterior pituitary and Negative feedback regulation
- ❖ Human Growth Hormone: Physiology and Pathophysiology

Hormones of Hypothalamus & Pituitary

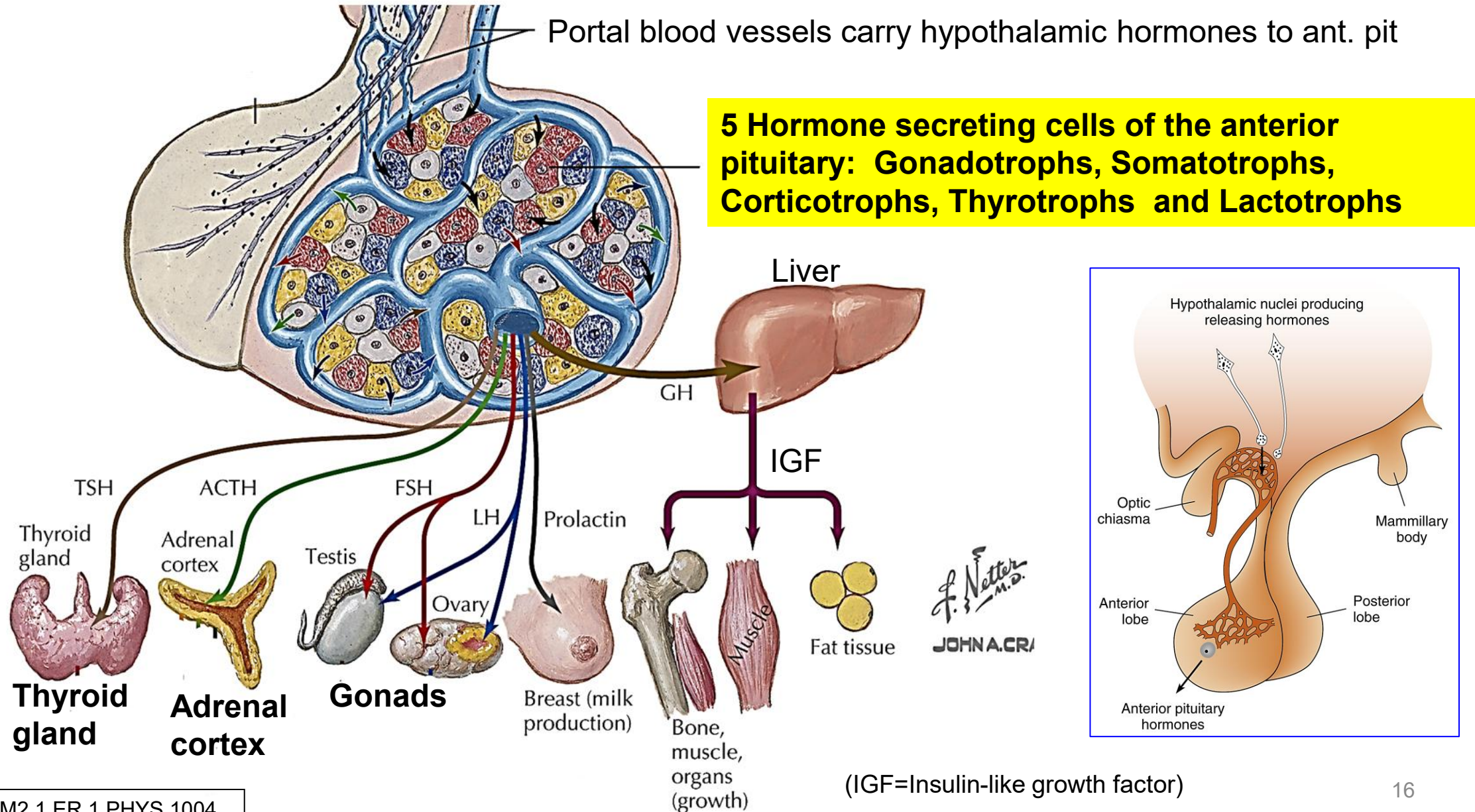


Hypothalamic neurons secrete Six Hypophysiotropic hormones



*GnRH is also known as LHRH; **SRIF=Somatotropin release inhibiting factor; ***Dopamine is also known as prolactin inhibiting hormone (PIH); TRH also stimulates prolactin secretion and therefore known as prolactin releasing hormone (PRH).

Anterior Pituitary Secretes Six Hormones



Anterior Pituitary Hormone Secreting Cells: Clinical correlate

Cell population:

Somatotrophs: Comprise the largest cell population (50%),

Corticotrophs: Make 15%-20%,

Gonadotrophs and Lactotrophs: Constitute 10%-15% each, and

Thyrotrophs: The least common cell type (10%)

Clinical correlate:

Tumors of anterior pituitary are called **pituitary adenomas**.

Adenomas may or may not secrete large quantity of one or more hormones.

The most common hypersecreting adenomas are Prolactin-secreting tumors (**Prolactinoma**).

Classification of Anterior Pituitary Hormones

3 Types based on their chemical structure

Clinical importance:
Hormones of the same family may cross-react when in excess.

Glycoprotein family:

- TSH, FSH, LH.
- Have α and β subunit chains
- α subunit is same in all but not β
- Hormone is active only when both subunits are together.

Pro-opiomelanocortin (POMC)
derivates: ACTH and β -Lipotropin

Growth Hormone family:

- GH and PRL.
- Have growth-promoting activity

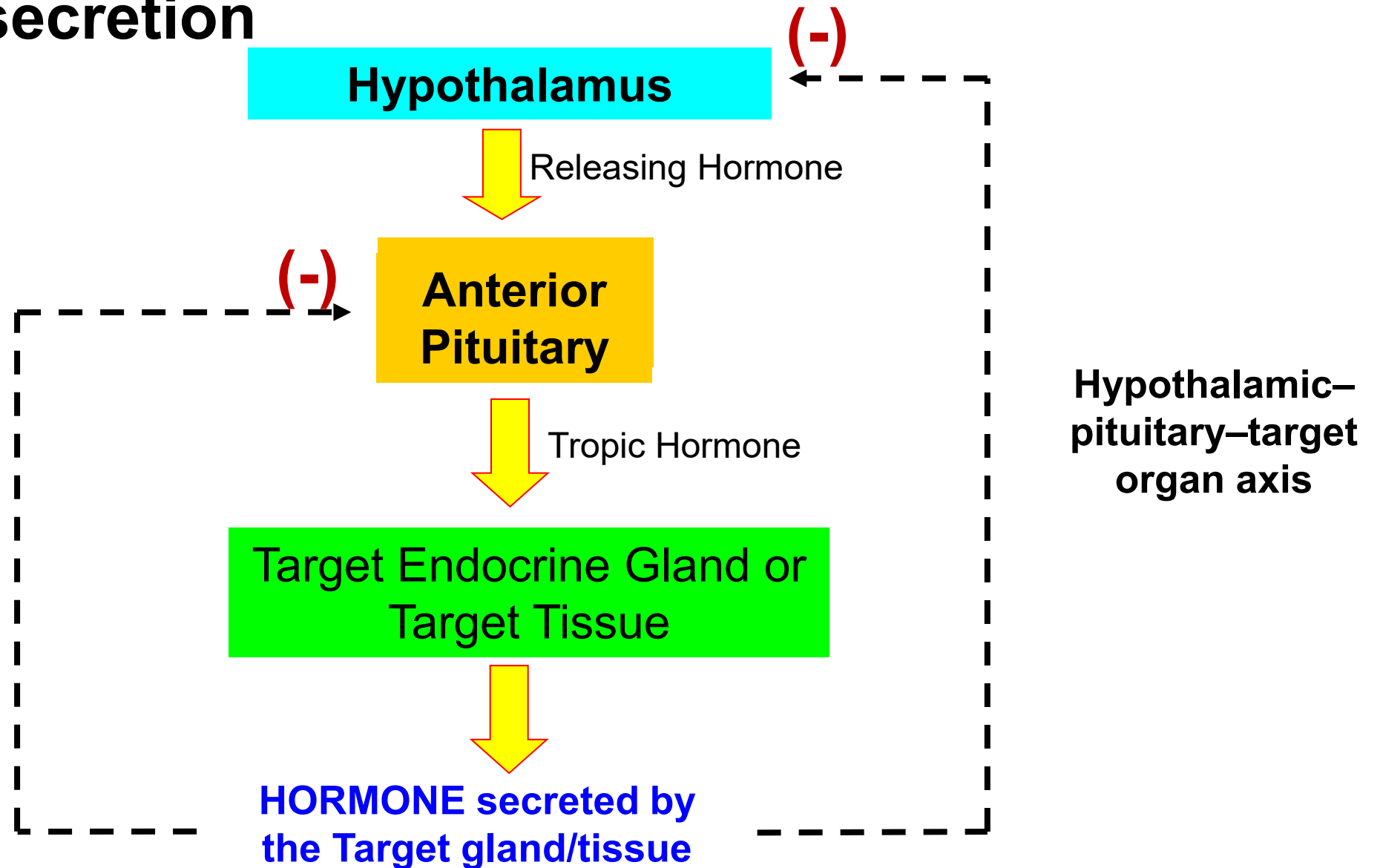
Placenta secretes:

1. Human Chorionic Gonadotropin (hCG) belongs to glycoprotein family.
2. Human Placental Lactogen (hPL) belongs to GH family.

Examples of clinical conditions in which excess of one hormone produces clinical manifestation of another hormone of the same family (cross-reactivity of hormones) due to structural similarity

- Patients having a tumor called choriocarcinoma have high blood levels of hCG. Because of structural similarity to TSH, this hormone (hCG) would stimulate thyroid gland and produce clinical features of hyperthyroidism.
- Both ACTH and Melanocyte stimulating hormone (MSH) are derived from a precursor molecule called, POMC. ACTH being a large polypeptide, contains certain amino acid sequence of MSH at its N-terminal end. Patients who have high blood levels of ACTH present with hyperpigmentation (darkening of skin) because of its melanocyte-stimulating activity. Diseases which cause high blood levels of ACTH like ACTH-secreting tumors and Addison disease present with hyperpigmentation of skin.

Negative Feed-back Control of the Anterior Pituitary Hormone secretion



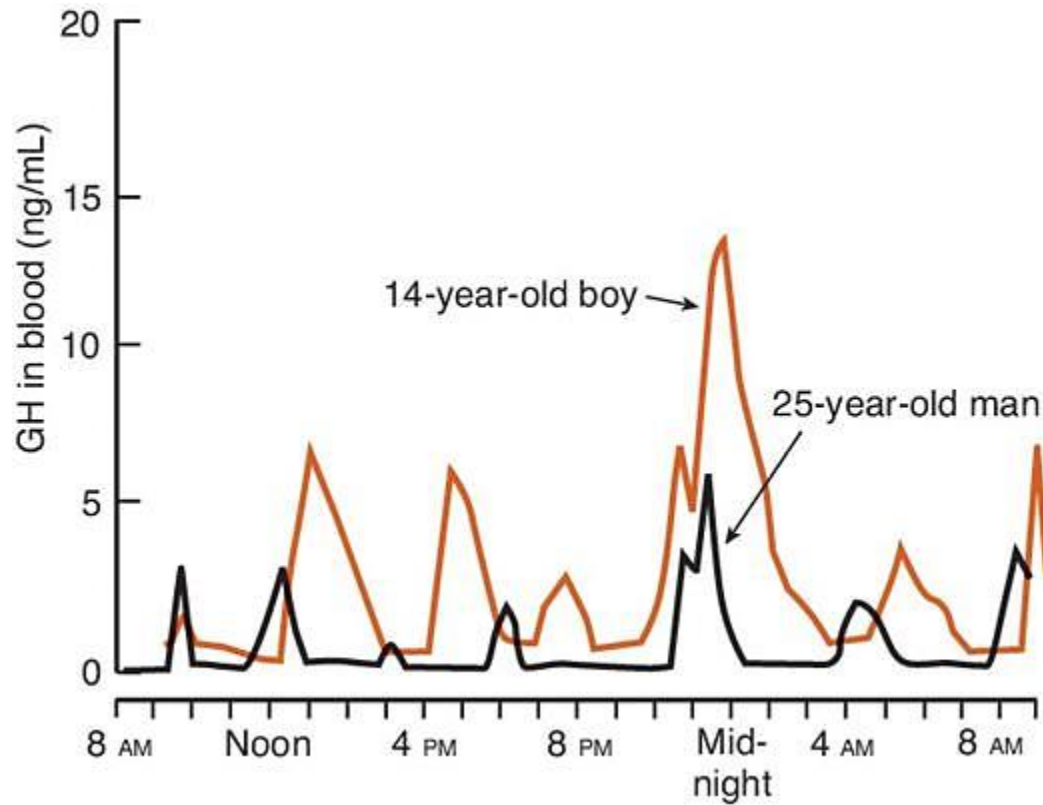
Lecture Outline

- ❖ Relationship between hypothalamus and pituitary gland
- ❖ Hormones of anterior pituitary and Negative feedback regulation
- ❖ Human Growth Hormone: Physiology and Pathophysiology

Growth hormone (GH): Features

- Secreted by somatotrophs in the anterior pituitary.
- GH is a 191 amino acid peptide.
- Has structural similarity to PRL and Human placental lactogen (hPL).
- **High concentrations** of GH can have PRL-like or lactogenic activity.
- Approximately 50% of the circulating growth hormone is in the bound form, bound to **growth hormone binding protein (GHBP)**. This provides a reservoir of the hormone to compensate for the wide fluctuations that occur in secretion during a day (see next slide).
- Secretion is high during **sleep (slow wave sleep), stress, and young age**; reduced in old age and obese individuals.
- GH acts **directly** and **indirectly** through another hormone, secreted from liver called Insulin-like Growth factor-1 (IGF-1).

Secretion of GH is Pulsatile

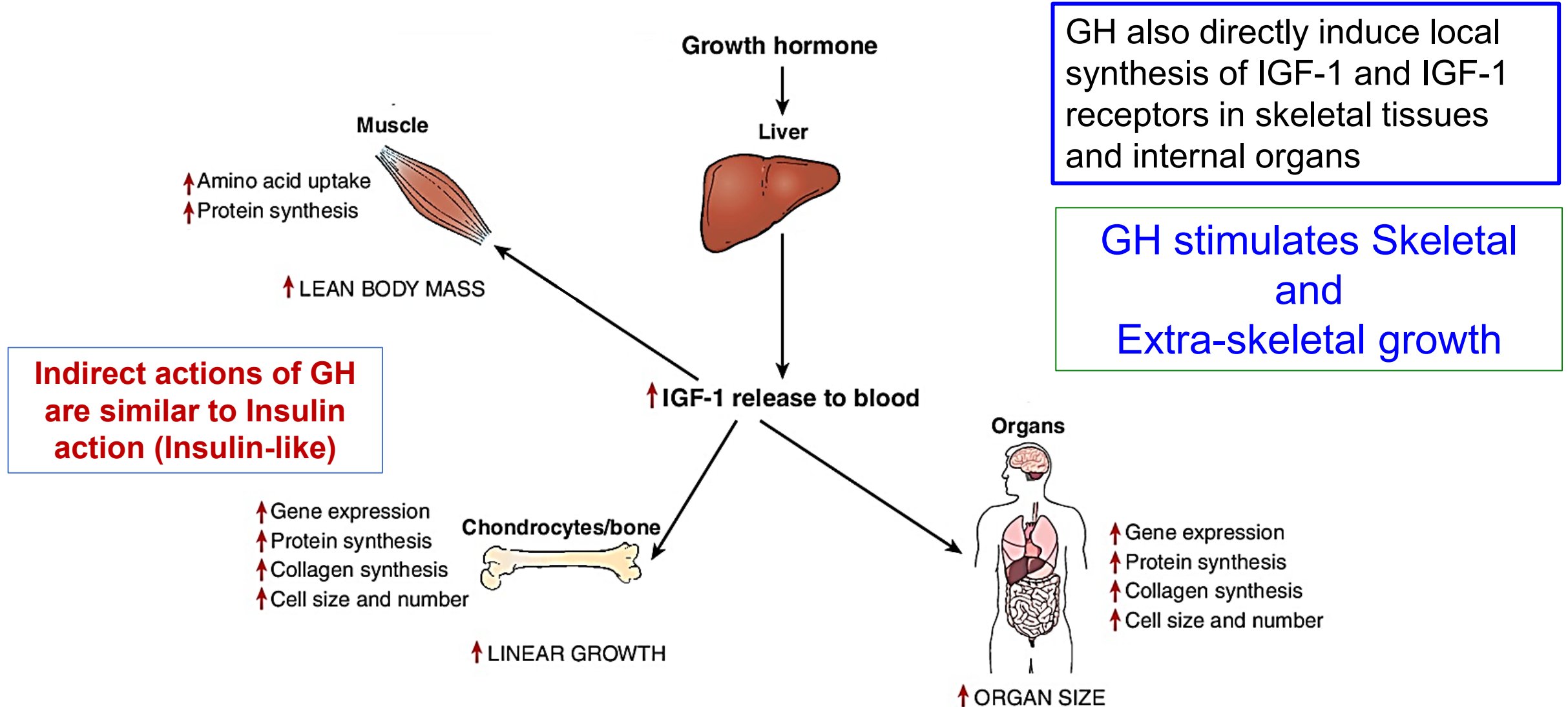


- Occurs in periodic bursts (**pulsatile**)
- Maximal peaks occur at night
- Usually during first hour after **deep sleep** (stage 3 & 4 of slow wave sleep)
- Bigger bursts during adolescence and smaller in old age and obese person.

Physiologic Actions of the GH

- GH has 2 major actions:
 - Growth** of skeletal and soft tissues and
 - Metabolic** on carbohydrate, lipid and protein metabolism.
- GH hormone exerts its actions by 2 ways:
 - Indirect** through another hormone called, insulin-like growth factor-1 (IGF-1), and
 - Direct** through GH-receptors in target tissue.

Indirect actions of the GH: IGF-1 mediates Growth Promoting effects of GH

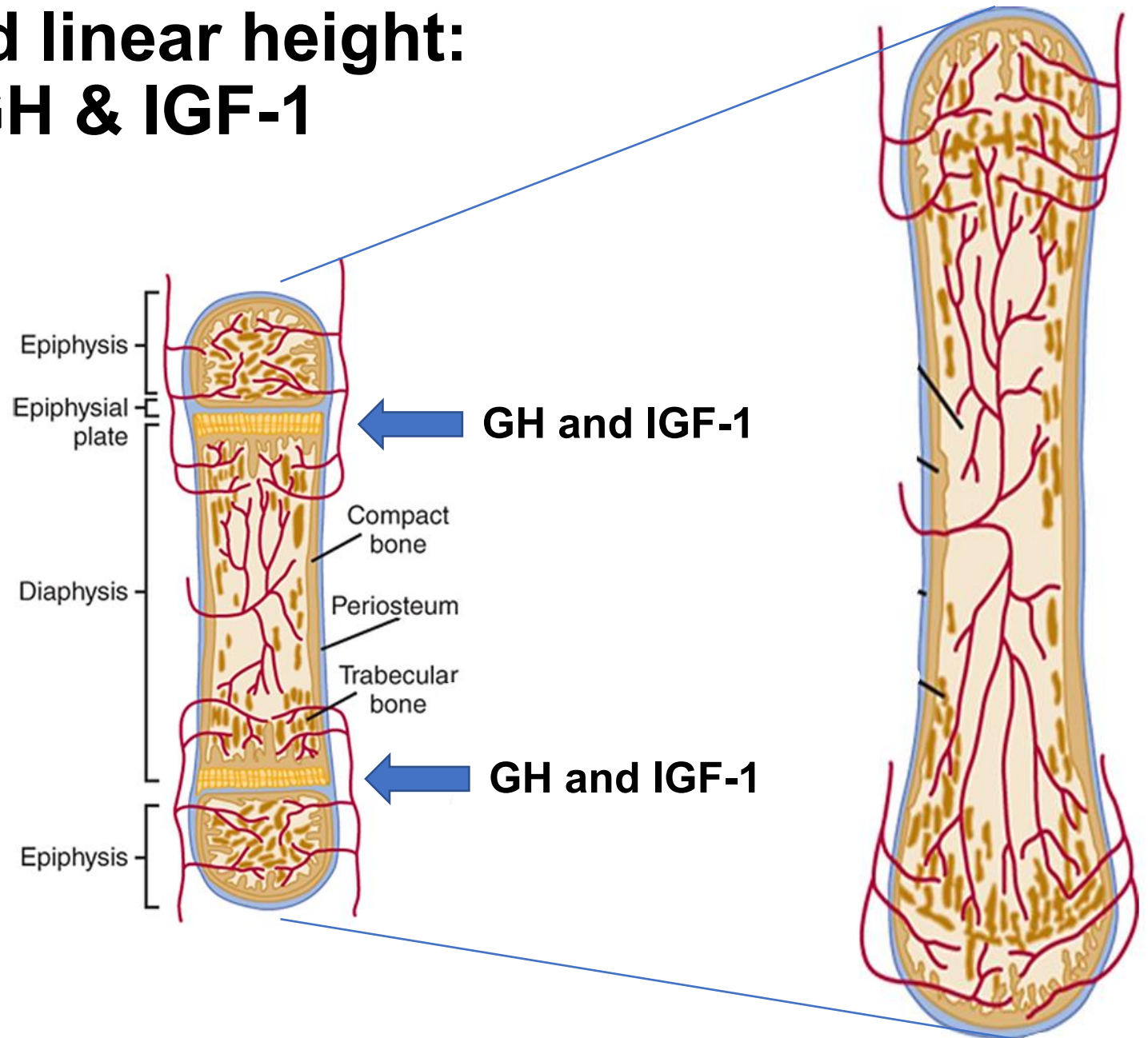


Insulin-like Growth Factor (IGF): Features

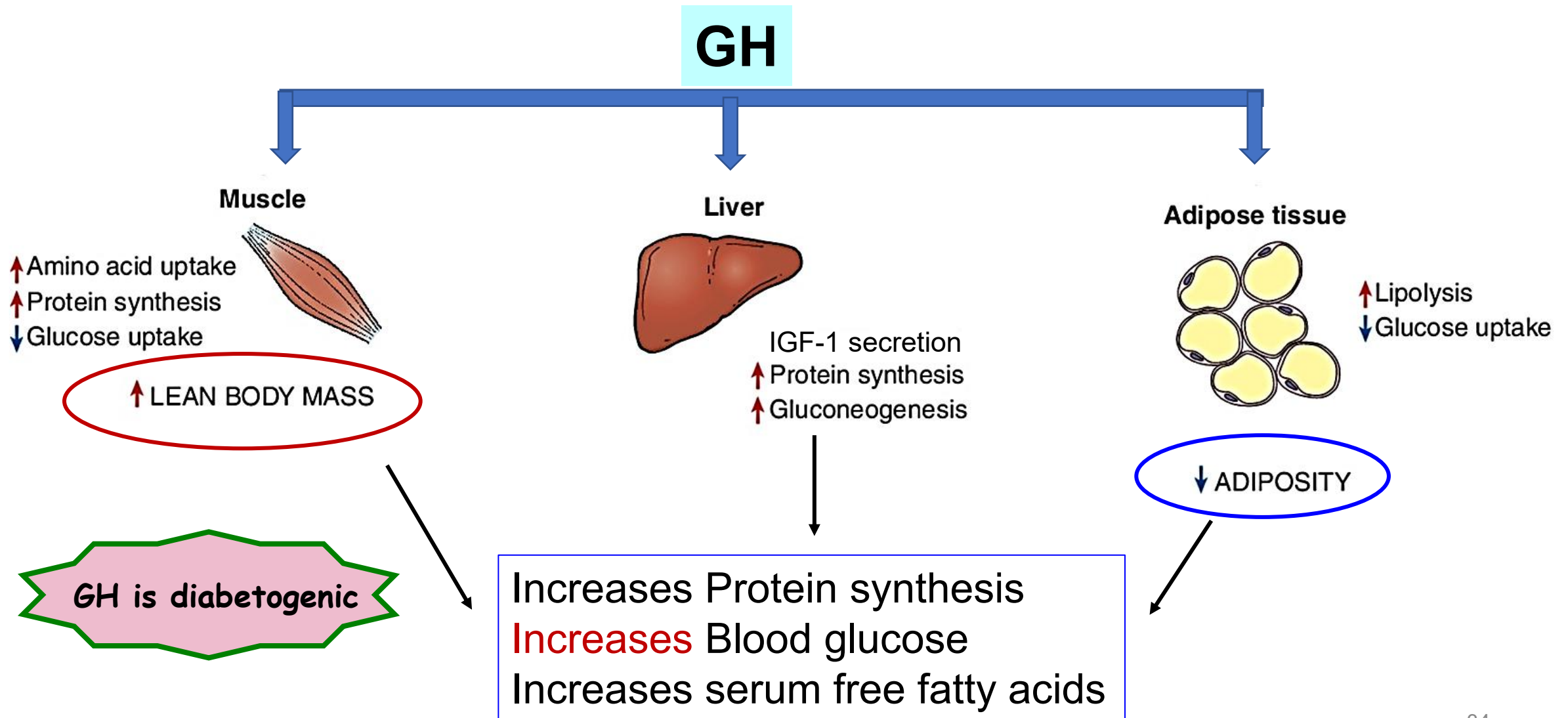
- There are 2 major subtypes of IGF: IGF-1 and IGF-2.
- IGF-1 (also known as somatomedin C) is the MOST important.
- IGFs are peptide hormones, structurally resemble insulin.
- IGFs are bound to binding proteins in the plasma, called IGF-binding proteins.
- IGF-1 is produced mainly in the **liver** and in other tissues in response to GH (GH → Liver → IGF-1 secretion).
- IGF-1 binds to the IGF-receptors in the target tissues that have tyrosine kinase activity.
- Defect in GH receptors results in decreased IGF-1 secretion by the liver that leads to dwarfism as in Laron dwarfism.

Skeletal Growth and linear height: Promoted by GH & IGF-1

- GH & IGF-1 increase width of the **epiphyseal plates**.
- Increase in the width with calcification increases the length of the long bone and linear height. **Thyroid hormones** has **permissive** action on GH in this process.
- In hypothyroid children GH action is impaired and results in dwarfism.



Direct actions of the GH: Regulates Metabolism in Liver, Adipose tissue and Muscles



GH has both Insulin-like and Anti-insulin Effects

All Indirect actions of GH are **Insulin-like** actions:

- ❖ Growth promotion
- ❖ Protein synthesis
- ❖ ↑Lean body mass
- ❖ Positive nitrogen balance

➡ **All “I”**
Indirect
Insulin-like
IGF-1 mediated

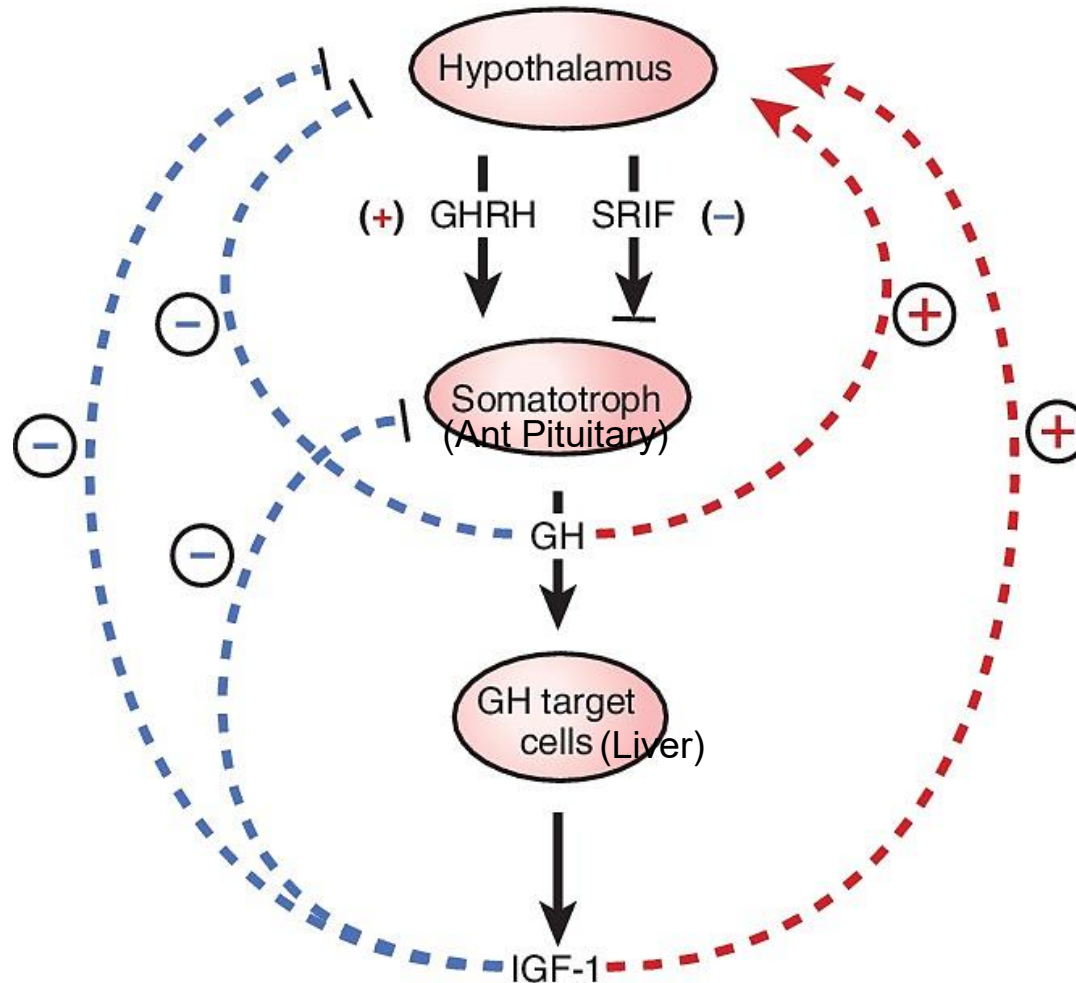
Most of Direct actions of GH are **anti-insulin** actions:

- ❖ Lipolysis
- ❖ ↑Blood glucose
- ❖ ↓Glucose uptake in tissues
- ❖ ↓ Insulin sensitivity

Clinical Correlate: Excess GH promotes skeletal and soft tissue growth; but **Diabetogenic because of insulin resistance!**

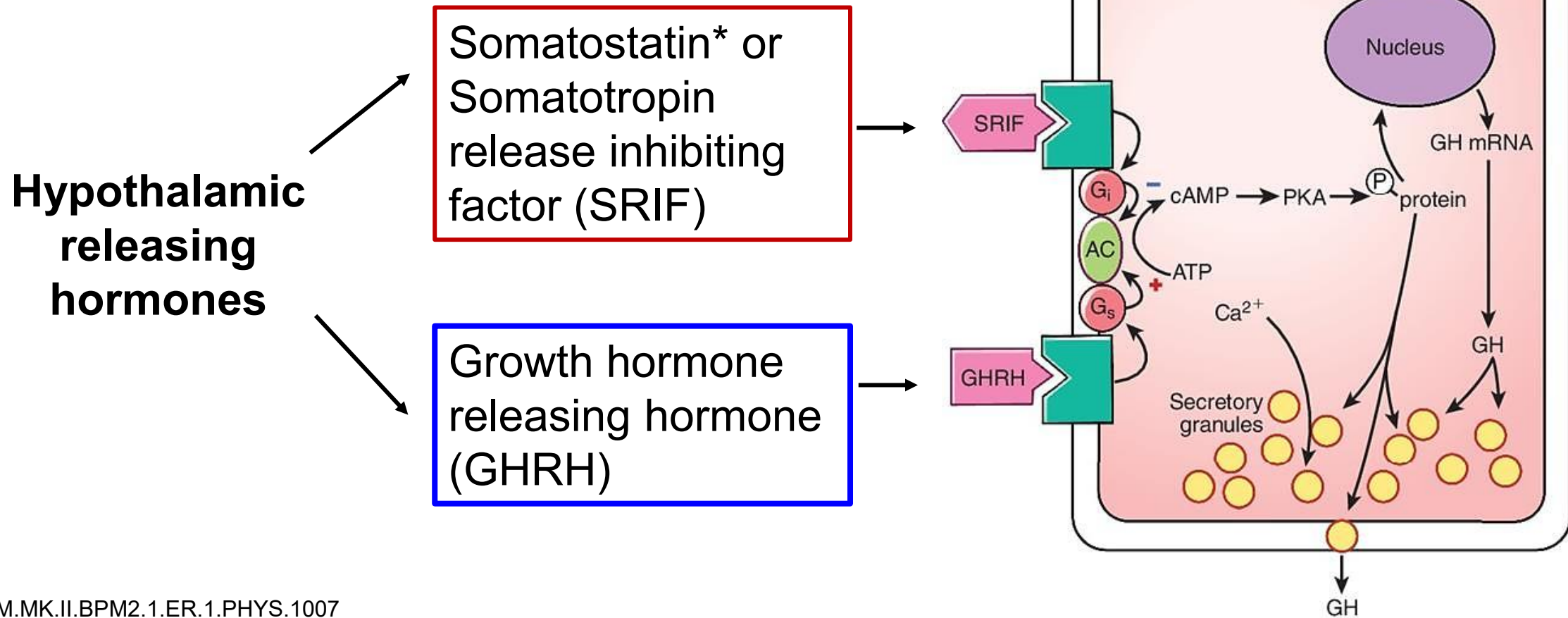
Regulation of Growth Hormone secretion: Hypothalamic–Pituitary–GH axis

IGF-1 inhibits
hypothalamic
GHRH (**Blue**
dotted lines)



IGF-1 stimulates
hypothalamic **SRIF**
(**Somatostatin**)
(**Red dotted lines**)

Growth hormone (GH) secretion is controlled by Hypothalamic Hormones



*Somatostatin is also known as Growth hormone inhibiting hormone (GHIH)

Regulation of Growth Hormone

- It is regulated by many different mechanisms.
- Growth hormone releasing hormone (GHRH) **stimulates** the release of GH.
- Somatotropin release inhibiting factor (SRIF, also known as Somatostatin) **inhibits** the release of GH.
- The above two hypothalamic hormones in turn are controlled by feedbacks from the circulating plasma levels of both GH and IGF-1:
 - GH and IGF-1 inhibit hypothalamic GHRH, thereby cause a decrease in GH secretion by the anterior pituitary;
 - GH and IGF-1 stimulate hypothalamic Somatostatin, thereby cause a decrease in GH secretion by the anterior pituitary.
- **Other additional factors** that control GH secretion include:
 - Factors that stimulate** GH secretion are: Stress, Ghrelin, Exercise, Hypoglycemia, and amino acids (arginine and leucine).
 - Factors that inhibit** GH secretion are: Hyperglycemia and Obesity.

Clinical Conditions associated with altered GH secretion

Effects of excess GH:

- ❖ Clinical features are because of exaggerated actions of the GH
- ❖ In children: Gigantism
- ❖ In adults: Acromegaly

Effects of decreased GH:

- ❖ Clinical features are because of too less GH in blood, or the target tissue is insensitive to GH
- ❖ In children: Dwarfism
- ❖ In adults: Adult GH deficiency

Excess GH secretion in children: Gigantism

- Excess GH secretion **before fusion of the epiphyseal plates** of the long bones.
- Rapid increase in height and weight.
- Major features: Large hands and feet, coarsening of the facial features with frontal bossing and prognathism (projection of the jaw), headaches, and excessive sweating.
- Other features: ↑Plasma GH and IGF-1, **↑Blood glucose**, and **↑BP**.
- Causes: Pituitary GH-secreting tumor, Excessive hypothalamic GHRH-secretion, and defective hypothalamic somatostatin secretion.

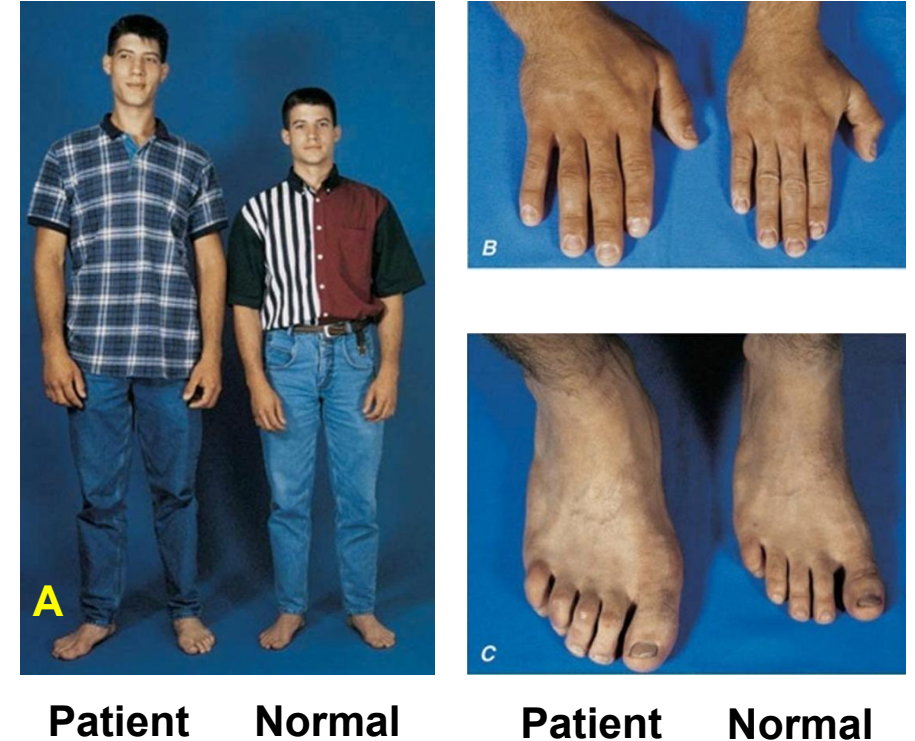


Fig. 22-year-old identical twins: A. Patient (left) and his normal twin brother (right). B & C: Note large hand and foot of the patient (left) compared with that of his normal brother (right). These changes began at the age of 13.

Excess GH secretion in adults: Acromegaly

- Excess GH secretion **after fusion** of the epiphyseal plates (in adults).
- Excessive skeletal, soft tissue and internal organ growths but no increase in height.
- **Acral bony overgrowth** results in frontal bossing, increased hand and foot size, mandibular enlargement with prognathism (projection of the jaw), and frontal bossing.
- Other features: Headaches, **↑BP**, excessive sweating, **↑plasma GH** and **IGF-1**, and **↑blood glucose**.

(Acral = Peripheral body parts, such as toes and fingers)



Fig. Facial aspect of a patient with acromegaly. A & B. Wide nose, bulging forehead and prominent cheek bone. C & D. Mandibular overgrowth (prognathism) and jaw malocclusion.

Excess GH secretion in adults: Acromegaly



Patient Normal



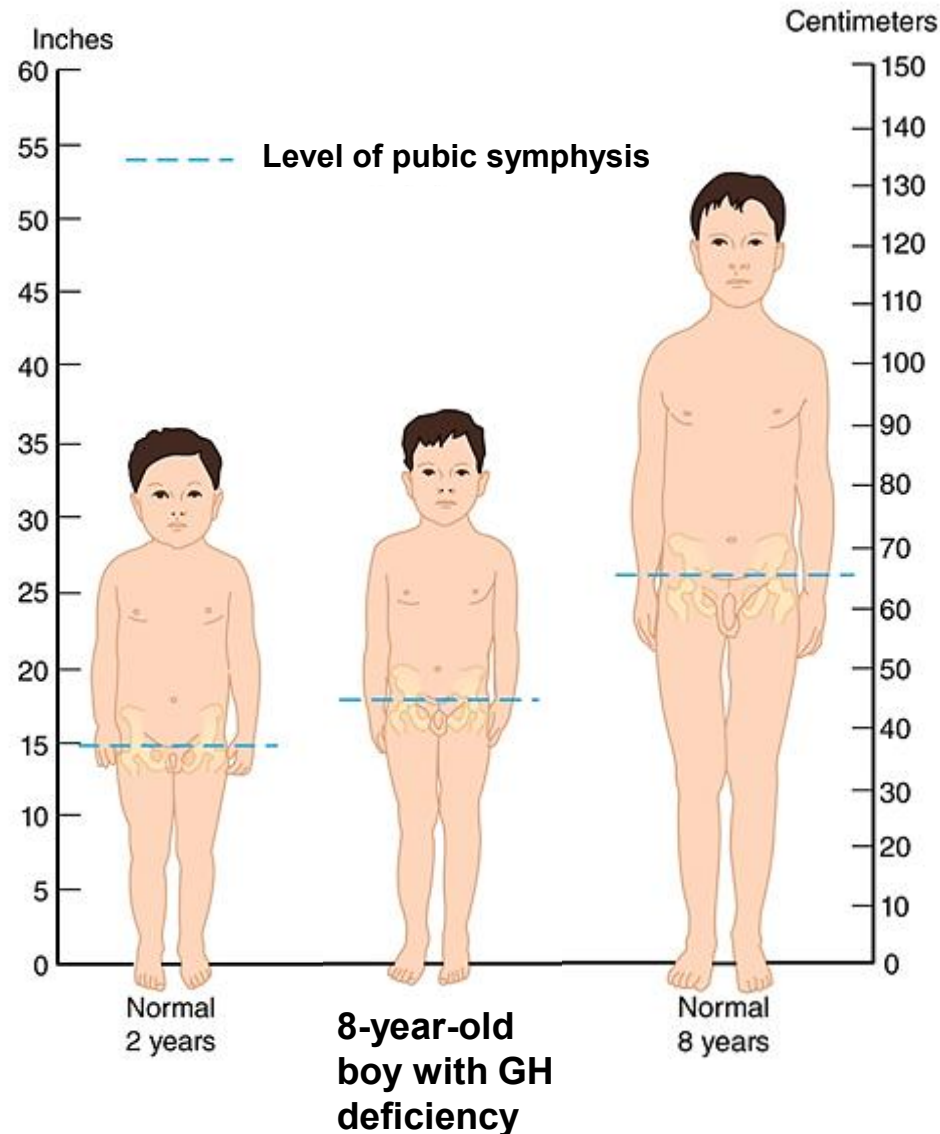
Acromegalic face



Normal Patient

(Acral = Peripheral body parts, such as toes and fingers)

GH deficiency in children: Dwarfism



Clinical features:

- Growth retardation,
- Short stature,
- Delayed bone age,
- Low blood glucose (Hypoglycemia).

More details of GH deficiency in the next lecture!

Other Recommended Readings

1. Section III: Endocrine & Reproductive Physiology. Ganong's Review of Medical Physiology, 26e. Kim E. Barrett, Susan M. Barman, Heddwen L. Brooks, Jason X.-J. Yuan.
2. General Principles of Endocrinology and Pituitary and Hypothalamic Hormones: Netter's Essential Physiology, Second Edition. Mulroney, Susan E., PhD.
3. Part 12: Endocrinology and Metabolism-Section 1. Endocrinology. Harrison's Principles of Internal Medicine, 21e. Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson.

Any Questions ?

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